Pantheon-Evolve 2 — A Redesign for Robust, Transparent, Cumulative Method Discovery

2026-07-10 · Pantheon-Evolve redesign for the Cell response-to-reviewers

Contents

Purpose. Address the two Cell reviewers' core objection to Pantheon-Evolve — that it is a competent-but-incremental re-assembly of existing LLM-guided code evolution whose headline gains may be metric-overfitting rather than real method invention — by changing *what is optimized and how validity is established*, not by tuning the existing loop. The redesign is also the algorithmic contribution that lets us claim novelty *beyond* SimpleTES, AutoScientists, and DeLM.

Audience: internal (rebuttal authoring + implementation). Method names/passages are written in English so they can be pasted into the manuscript/response-to-reviewers.

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0. TL;DR

The current Pantheon-Evolve maximizes a **fixed, human-authored scalar** `f = Σ wᵢ·metricᵢ (+ LLM reviewer)` and then reports components of *that same scalar*. Every serious reviewer criticism is a symptom of this one design choice (Goodhart's law), plus a transparency gap. Both are structural, not presentational.

Pantheon-Evolve 2 (PE2) replaces the fixed scalar with a **minimax co-evolution between a population of candidate methods and a population of adversarial falsifiers**, gated by a **learned validity model** that separates real biological signal from known artifact classes, with every accepted improvement distilled into a **transparent, reusable, cross-domain "method-gene"** carrying a **generalization certificate.** A **learnable search policy** makes the system improve at *how* it evolves across runs and turns the reproducibility question into a reported distribution.

The one-line pitch for the rebuttal:

> *We no longer ask "did the score go up?"; we ask "can an adversary find any held-out study, technology, or orthogonal biological readout where this method collapses, and is the gain explained by a known artifact?" A method ships only when the answer is no, with a certificate.*

This is Goodhart-resistant **by construction**, and it is precisely the gap that SimpleTES (names evaluator fidelity as its central limitation), AutoScientists (single-objective, no learned policy), and DeLM (no evolution, verifies *evidence support* but not *biological reality* or *OOD generalization*) all leave open.

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1. Diagnosis — one root cause behind the reviews

1.1 What both reviewers actually said about Pantheon-Evolve

#	Reviewer point	Root issue
R1-2f, R2-8, R2-10	Objective = reported metric → **circular / metric-overfitting**; "beautiful not real"; batch-mixing improvable by **overcorrection**	Fixed proxy objective
R2-10	Objective is a **hand-weighted scalar** (integration quality + bio-conservation + speed + convergence, with function hints) → "guided optimization, not method invention"	Human-designed objective

#	Reviewer point	Root issue
R2-10, R1-min1	Train/val/test split of the **same cells** ≠ generalization across studies/tissues/technologies	No OOD protocol
R1-2h	**Black box**: user sees score rise, not *what changed and why*	No mechanistic output
R1-3	**Single run**, no variance across seeds, no failure distribution; Harmony/BBKNN "more modest" unexplained	No distribution-al reporting
R1-4, R2-10	"Super-human" vs **Scanorama-2019**, not vs an expert human + Claude Code in 2026 with the same harness	Wrong counterfactual
R2-9	Resembles existing LLM genetic programming; analyzer/mutator/MAP-Elites/islands/MTP/LLM-reviewer **asserted, not ablated** as decisive	No ablation of novelty
R1-5	Multiple evolved variants — **no mechanism to pick** the right one	No variant router
R1-11, R2-7	Cost = API only; no time/robustness/token/failure-rate reporting	No practical-cost accounting

Nine complaints, **two root causes**: (A) a fixed human proxy objective the search Goodharts against; (B) opacity + no distribution. PE2 attacks A with Primitives 1–3, B with Primitives 2 & 4 and the evaluation protocol.

1.2 What the code read confirms (so we don't over-defend the current build)

The current implementation is honestly an AlphaEvolve/OpenEvolve re-implementation, and several of its "novel" parts are inert or mis-wired — Reviewer 2's "novelty is incremental" is technically accurate about the shipped code:

- **Not a genetic algorithm** — no crossover; every child derives from one parent (`program.py`: singular `parent_id`). The "genetic-algorithm-driven" label in `README.md:41` is a misnomer; the effective algorithm is a single-parent LLM hill-climber.
- **MAP-Elites descriptors are auto-set to the fitness metrics themselves** (`team.py:632–643` takes the first two keys of `fitness_weights`), so the quality-diversity grid collapses onto the objective axes — it illuminates nothing orthogonal.
- **Population never pruned / bounded** (`population_size` in `config.py:41` is unused; nothing is ever deleted from `database.programs`); the MAP-Elites `accepted` flag is cosmetic.
- **Islands are inert** — parent selection passes `island_id=None`; migration copies only id references, never isolates genomes, and is skipped entirely in the parallel path.
- **Cascade evaluation is dead config** (`cascade_evaluation` / `cascade_thresholds` defined, serialized, never implemented).
- **Fitness is a single non-stationary scalar** — weighted sum (`function_weight=0.8`, `llm_weight=0.2`) with min-max ranges that expand over the run, so scores aren't comparable across time; no Pareto, no novelty term.

Strategic implication: we should **not** defend the current machinery as-is. We should present PE2 as a *new formulation* that (i) makes the previously-inert QD machinery actually do something (real orthogonal descriptors, bounded archive), and (ii) adds the genuinely novel minimax + validity + method-gene layer that no baseline has.

2. Pantheon-Evolve 2 in one picture

```
                     ┌────────────────────── PE2 co-evolution ──────────────────────┐
  method-gene        │  Population P of METHODS          Population A of FALSIFIERS  │
  LIBRARY   ◄────────┤  (programs)                       (held-out studies / techs / │
  (typed motifs,     │       │                            metrics / stress transforms /
   preconditions,    │       │  propose (policy π)        "trap" datasets)          │
   certificates)     │       ▼                                 │                     │
       ▲             │  child method  ────────────────►  scored on WORST-CASE across A
       │             │       │         fitness(m) = CVaR_χ∈A [ V-gated held-out score ]
   ABSTRACT &        │       │                                 ▲                     │
   CERTIFY  ─────────┤       │         fitness(χ) = regret χ exposes on elite methods
       ▲             │       ▼                                 │                     │
       │             │  VALIDITY MODEL V  ── flags artifact (overcorrection / boundary
  learnable          │  sharpening / imputation-imposed structure) → discounts "fake" gain
  POLICY π  ◄────────┤                                                                │
  (cross-run credit) └────────────────────────────────────────────────────────────────┘
```

Four primitives (each maps to reviewer criticisms and to a gap in the three baselines):

1. **Adversarial Evaluator Co-Evolution** — methods vs falsifiers, minimax / distributionally robust fitness. *Kills circular eval & overcorrection.*
2. **Mechanistic Method-Gene Library** — every accepted change becomes a transparent, typed, reusable motif with a precondition. *Kills the black box; makes evolution cumulative and cross-domain.*
3. **Objective Elicitation + Validity Model** — no hand-weighted scalar (Pareto), plus a learned real-vs-artifact gate. *Kills human-designed objectives & "beautiful-not-real."*
4. **Learnable Search Policy + real Quality-Diversity** — the strategy is learned across runs; fix the inert QD. *Answers reproducibility (raise the floor / shrink variance) & the "not a real GA / not ablated" charge.*

Plus a **variant auto-router** (answers R1-5) and an **anti-circular evaluation protocol** (P1–P6) that is itself the rebuttal.

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3. Primitive 1 — Adversarial Evaluator Co-Evolution (the anti-Goodhart core)

3.1 Formalism

Replace the fixed objective `f(m)` with a **two-population minimax game.**

- **Methods** `m ∈ P` — programs (as today).
- **Falsifiers** `χ ∈ A` — each a challenge tuple `χ = (D_χ, μ_χ, τ_χ)` where `D_χ` is a *held-out* data condition (a study / technology / tissue / perturbation the method never trained on, or a **trap** dataset with known structure), `μ_χ` is a *metric family* (possibly one the method never optimized against), and `τ_χ` is an optional *stress transform* (dropout, batch imbalance, orientation permutation, label shuffle on nuisance axes).

Method fitness = worst-case (or CVaR at level α) validity-gated score over the current falsifier ensemble:

```
Fit(m) = CVaR_{χ ~ A, α} [ V(m, χ) · score(m; D_χ, μ_χ, τ_χ) ]
```

where `V ∈ [0,1]` is the validity model (Primitive 3) that discounts artifactual gains. CVaR (average of the worst α-fraction) rather than a hard min gives a smoother, less brittle objective while preserving robustness.

Falsifier fitness = the *regret* / generalization gap the falsifier exposes on the current elite methods `E ⊂ P`:

```
Fit(χ) = E_{m ∈ E} [ score_in−distribution(m) − V(m,χ)·score(m; D_χ,μ_χ,τ_χ) ]
```

i.e. a falsifier is rewarded for finding a held-out condition where a method that *looked* good in-distribution actually collapses (or where the validity model catches an artifact).

3.2 Why this is Goodhart-resistant (the rigor hook)

This is **population-based distributionally-robust optimization (DRO)** where the uncertainty set is *learned and expanded by an adversary*, connecting to GAN-style co-evolution and to open-ended learning (POET / minimal-criterion coevolution).

- Under a **fixed** evaluator, `argmax_m f(m)` is by definition the *best proxy-exploiter* on the training distribution — exactly the reviewer's "optimize the benchmark" failure.
- Under the **minimax**, a method can only win by having a *small generalization gap over the falsifier-reachable distribution*. If a method improves an in-distribution metric via overcorrection, the bio-null trap falsifier (below) drives `Fit(m)` down; if it overfits one study, the leave-study-out falsifier does. **The only stable way to win is genuine transfer.**

Proposition (stated plainly for the paper): *for any method `m`, `Fit(m)` upper-bounds `m`'s worst-case held-out performance over the class of conditions the falsifier population can realize; therefore a high co-evolutionary fitness certifies bounded generalization gap, whereas a high fixed-scalar score certifies nothing beyond the training split.*

3.3 The falsifier classes (this is where domain expertise enters — as *tests*, not weights)

Instead of the human writing objective *weights*, the human (or the system, ratified by the validity model) contributes **falsifier generators** — biologically-motivated ways to try to break a method. Generic classes, instantiated per domain in §7:

- **Leave-condition-out**: hold out a study, a technology (10x vs Smart-seq vs MERFISH), a tissue, a donor, a perturbation, a time point.
- **Orthogonal-readout**: score on a biological objective the method never optimized (rare-cell recovery, spatial-domain preservation, ligand–receptor program, DE preservation, held-out *measured* genes).
- **Trap datasets** — the reviewers' own failure modes weaponized as fitness:
    - *bio-null trap*: batches sharing no real biology → any "mixing" gain is overcorrection.
    - *orientation/artifact trap*: geometry with a permuted axis → any recovered "gradient" that survives is an artifact (directly operationalizes R2-11's Cer1-boundary circularity worry).
    - *imputation-null trap*: reference structure withheld → any recovered pattern is imposed.
- **Stress transforms**: subsampling, batch-size imbalance, dropout inflation, label noise on nuisance covariates.

Co-evolution means the falsifier population **grows toward whatever currently-elite methods are weakest at** — the search for counter-examples is automated, not hand-listed.

3.4 Mitigating co-evolutionary pathologies (be honest in the paper)

Co-evolution can cycle or "disengage." Standard, citable mitigations we adopt:

- **Falsifier hall-of-fame** + **quality-diversity on falsifiers** (keep a diverse archive of hard challenges so the pressure doesn't collapse to one trick).
- **Minimal-criterion coevolution** (POET): a falsifier is admitted only if it is *solvable by at least one* current method and *unsolved by the elite* — keeps challenges in the learnable band.
- **Gradient of difficulty**: report an explicit "falsifier difficulty ladder" so reviewers see the pressure is real and increasing.

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4. Primitive 2 — The Mechanistic Method-Gene Library (anti-black-box, cumulative, cross-domain)

4.1 Every accepted mutation ships a falsifiable mechanistic claim

Today the summarizer post-hoc classifies a diff into `{direction, category, is_algorithmic}`. PE2 requires more: a child is admitted only with a **method-gene** record:

```
MethodGene {
  name:         "adaptive local-density kNN kernel"      # human-readable motif
  code_delta:   <the diff that introduces it>
  claim:        "replaces fixed bandwidth with per-cell density-adaptive bandwidth,
                 reducing over-smoothing in dense manifolds"          # falsifiable
  precondition: "helps when local density varies > k-fold across the manifold"  # WHEN it helps
  typed_params: { k: int, alpha: float ∈ [0,1] }
  ablation:     <auto-run A/B: method with vs without this gene on the falsifier set>
  effect:       { held_out_delta: +0.06 ± 0.02, artifact_p: 0.31 }    # measured, certified
}
```

The **ablation is run by the system** (method with vs without the gene, on held-out falsifiers) so the claim is *tested*, not asserted. This directly answers R1-2h (transparency: the user sees what changed *and the causal test that it helps and when*) and R2-9 (the analyzer/mutator contribution is now *demonstrated* per-gene, not asserted).

4.2 The library makes evolution cumulative and cross-domain — the real "discovery" claim

Accepted method-genes accumulate into a typed **library / grammar** (DreamCoder-style library learning). This buys three things no baseline has:

1. **Transparency** — the evolution trace becomes a readable list of named, precondition-tagged motifs, not opaque diffs.
2. **Cumulativity** — later runs *compose* known genes instead of re-deriving them (fewer evaluations to reach a given robustness → answers cost, R1-11/R2-7).
3. **Cross-domain transfer** — a gene discovered on **batch correction** ("adaptive local-density kernel") becomes a *candidate operator* for **3D kernel regression** and **gene-panel graph construction**. Demonstrating that a motif discovered in one task provably improves a held-out *different* task is a concrete, checkable "method invention" result — the thing Reviewer 2 says is missing ("autonomous scientific method invention, not guided optimization").

This reframes the deliverable from "*we beat Scanorama on this metric*" to "*we discovered a transferable, mechanistically-explained, precondition-tagged algorithmic motif and certified when it helps.*" That is a method-development contribution a Cell method-dev reviewer can respect.

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5. Primitive 3 — Objective Elicitation + Validity Model (kill hand-weights; detect eval-hacking)

5.1 No scalarization — Pareto front

Replace `f = Σ wᵢ·metricᵢ` with **multi-objective non-dominated sorting** (NSGA-II style) over the objective vector `(integration, bio-conservation, robustness, cost, …)`. There are **no hand-tuned weights**, so R2-10's "human-designed objective weights" objection evaporates. The system reports a **Pareto front** and its trade-off structure; the variant router (§6) picks a front member for a given dataset. (AutoScientists explicitly lists multi-objective as *future work* — this is a concrete place we exceed it.)

5.2 The validity model V — real signal vs known artifact

`V(m, χ) ∈ [0,1]` is a learned model that predicts whether a metric improvement reflects real biological signal or a **known artifact class**, trained on data where the answer is knowable:

- **simulations** with known batch/mixing structure (over-mixing is ground-truth detectable);
- **held-out measured genes** (imputation that "predicts" masked measured genes = real; that invents structure = artifact);
- **orthogonal assays** (targeted spatial validation, independent modality);
- **negative controls** (permuted axes / null batches).

`V` enters fitness as a **gate/penalty**: a metric gain that `V` flags as overcorrection, boundary-sharpening, or imputation-imposed structure is *discounted*. This is the "co-evolution detects reward-hacking / fixes the evaluator" robustness story — and it operationalizes Reviewer 2's *specific* named failure modes (overcorrection; Silhouette an "incomplete safeguard"; imputation imposing reference structure; Cer1-boundary circularity) as first-class, penalized terms. **None of SimpleTES / AutoScientists / DeLM has this** — SimpleTES names reward-hacking as an *unsolved* limitation; DeLM verifies *evidence support* (does the claim match its source) but not *biological reality vs artifact*.

5.3 On objective autonomy — claim it carefully

Reviewers punished autonomy overclaims. Precise stance: **the system proposes candidate objective terms and falsifiers; the validity model + human ratify them.** We claim "the human specifies the *question* and the *sources of validity ground truth*; the system discovers the *method* and *the conditions under which it holds*." Mark this boundary explicitly (also answers R1-7 human-vs-agent boundary).

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6. Primitive 4 — Learnable Search Policy + real Quality-Diversity

6.1 Learn *how to evolve* across runs

The proposal policy `π` (which method-gene to apply, explore vs exploit, which parent, which falsifier to target) is **learned across runs** via trajectory-level credit assignment (SimpleTES-style IRFT/RLVR, or an accumulating ACE-playbook of "edit E in context C yielded held-out gain G"). This makes the *strategy* the object that generalizes — the "learnable, not frozen search" thesis — and it is the natural home for the reproducibility answer:

> *Over rounds, the learned policy should **raise the median and worst-case** outcome and **shrink variance**, not just the best run.*

This is exactly the distribution R1-3 asked for, reframed as a *result*: the learned-policy arm Pareto-dominates the fixed-search arm across seeds, not anecdotally.

6.2 Make the quality-diversity real (fix the inert machinery)

- **Behavior descriptors orthogonal to fitness** (the current bug): descriptors = *method-gene composition*, *compute cost*, *robustness profile shape across falsifiers* — none of which is the objective. Now the MAP-Elites grid illuminates genuinely different *kinds* of method.
- **Bounded, pruned archive** with content-hash dedup (the `content_hash()` that currently exists but is unused), so memory and $O(N^2)$ bookkeeping stop growing without bound.
- **Replace inert islands** with the method/falsifier predator–prey structure of Primitive 1 (which *is* a real multi-population model), or fix island genome-isolation if kept.

6.3 Reinstate cascade evaluation (dead config → cost lever)

The minimax raises evaluation count (more falsifiers = more evals). Pay for it with the **cascade evaluation that is currently defined-but-dead**: cheap falsifiers first, promote only survivors to expensive held-out assays; and adopt SimpleTES's "scale width not depth" and cheaper open-source generators (gpt-oss beat frontier models in their study). Report the resulting cost table (§9) — this answers R1-11/R2-7 head-on.

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7. Variant auto-selection (answers R1-5 and "how does the agent contribute")

The Pareto front + validity model + each variant's **falsifier profile** gives a principled router: given a user's dataset `D_user`, find the falsifier `χ*` whose condition is most similar to `D_user`, and recommend the front member with the best certified `V·score` on `χ*`. This is a "which method for which regime" **router**, so the system *navigates between variants* instead of dumping alternatives on the user (R1-5's exact ask), and it makes the **agentic layer load-bearing** rather than a wrapper (R1-2 final point, R2-8): the agent's job is *method selection and certification under the user's regime*, which no fixed pipeline does.

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8. The anti-circular evaluation protocol (P1–P6) — this *is* the rebuttal

Ship the redesign with a protocol that structurally cannot be accused of circularity. State it once, apply it to all four domains.

- **P1 — Objective/report disjointness.** Metrics used *inside* the evolution objective are disjoint from metrics *reported*. Enforced and logged. (Answers R1-2f, R2-8.)
- **P2 — Leave-condition-out generalization.** Evolve on `C_train`; report only on held-out studies/technologies/tissues/perturbations `C_test` never seen in the loop. Splitting the *same* cells is banned. (Answers R2-10, R1-min1.)

- **P3 — Adversarial validity.** Every headline gain must survive the falsifier ensemble + `V`; report an **artifact-rejection p-value** per claim. (Answers "beautiful not real", R1-2g.)
- **P4 — Right counterfactual.** Baseline arms under *identical* harness / budget / model backbone: (a) **expert human, one week**; (b) **Claude Code**; (c) **AlphaEvolve / OpenEvolve / AIDE** (the code-evolution systems R2-6 named). "Beyond human baseline" replaces "super-human" (R1-4, R2-10). (Answers R1-4, R2-6, R2-10.)
- **P5 — Distributional reporting.** ≥ N seeds; report **median / worst / IQR** and the **failure rate** (fraction of runs producing ≤ the input method), with the **dashed baseline line** the reviewer literally asked for. (Answers R1-3.)
- **P6 — Certificate.** Each shipped method carries `{mechanistic claim, held-out interval, artifact p-value, precondition}` — a single reviewable object. (Answers R1-2h.)

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9. Application to the four target tasks

For each: *what evolves*, *falsifier design*, *validity/trap*, *held-out claim*, *counterfactual*. The template is identical (that uniformity is itself a contribution — one robust method-discovery algorithm, four domains), which is what makes the platform claim credible rather than a grab-bag.

9.1 Batch correction / data integration (flagship — highest-leverage rebuttal)

- **Evolves:** the integration algorithm (Scanorama / Harmony / BBKNN and *de novo* variants).
- **Falsifiers:** leave-one-study-out, leave-one-technology-out; **bio-null trap** (batches with no shared biology → mixing gain = overcorrection); orthogonal held-out biology (rare-state recovery, DE preservation).
- **Validity:** `V` penalizes iLISI/kBET mixing gains that co-occur with loss of held-out biological signal (directly answers "mixing improvable by overcorrection; Silhouette incomplete safeguard").
- **Held-out claim:** improved integration on **studies/technologies never in the loop**, not a same-cells split (answers R2-10 head-on).
- **Counterfactual:** expert-week + Claude Code + AlphaEvolve/OpenEvolve/AIDE, identical budget; distribution over ≥10 seeds with a dashed "original Scanorama" baseline (answers R1-3, R1-4).
- **Also resolves R1-3's asymmetry:** Harmony/BBKNN being "more modest" becomes a *result* (their robustness ceiling is already near the certified front — the certificate shows *why*), not an embarrassment.

9.2 Gene panel design

- **Evolves:** the *selection algorithm* (not "run a fixed RL"). Method-genes: consensus-fill rule, adaptive panel-size, Pareto-vs-RL routing.
- **Falsifiers:** held-out tissue/technology + **orthogonal objectives the panel never optimized** — spatial-domain preservation, rare-cell-state detection, ligand–receptor program recovery (exactly R2-8's list), ideally against a real targeted assay.
- **Anti-circular:** objective uses ARI-family on `C_train`; **report** on the orthogonal held-out biology (answers R1-2f/R2-8 "circular ARI").
- **Agentic contribution made concrete:** the agent *discovers the selection rule and its precondition* and *routes* between RL/Pareto/consensus by regime — answering R1-1's "the

multi-agent system only seeds a list" and R1-5's "which variant."

9.3 3D reconstruction (embryo spatial landscape)

- **Evolves:** the reconstruction / registration / imputation algorithm.
- **Falsifiers/traps: orientation-artifact trap** — withhold the Cer1-defined boundary / permute embryo orientation and test whether the recovered "biological gradient" survives; **cross-embryo reproducibility** — must hold across all 6 embryos, not one representative.
- **Validity:** `V` penalizes gradients explainable by smoothing/orientation artifact — this literally turns Reviewer 2's circularity objection (R2-11) into the fitness function.
- **Held-out claim:** the recovered proximal–distal axis is certified *not* an orientation/ reconstruction artifact, reproducibly across embryos.

9.4 Virtual-cell modeling downstream task

- **Evolves:** the downstream method (perturbation-effect prediction / GRN inference / model routing).
- **Falsifiers:** unseen perturbations + unseen cell lines (e.g. Essential Perturb-seq transfer K562/RPE1 → HepG2/Jurkat); held-out *measured* genes; orthogonal assays.
- **Fixes vcRouter "tautology" (R2-14):** the router's ground truth becomes **real downstream task performance on held-out biology**, not registry metadata — so routing accuracy measures "did it pick the model that actually performs best here," not "can it retrieve metadata."
- **Held-out claim:** predicted perturbation effects validated on measured held-out genes / orthogonal assay, with uncertainty (answers R2-15 "speculative overlay").

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10. Positioning vs the three baselines (+ the code-evolution family)

System	Search unit	Objective	Evaluator	Learned policy	Multi-obj	Validity / artifact	Transparency
AlphaEvolve / OpenEvolve / AIDE	program (single pop)	fixed scalar	trusted static	no	no	no	opaque diffs
SimpleTES	proposal loop (C,L,K)	task score	static (names **reward-hacking as its limit**)	**yes** (IRFT)	no	no	trajectory only
AutoScientists	self-org team	single metric	2σ noise-gate + seed re-run	no (reactive)	**future work**	no (only significance)	report/model-card
DeLM	shared-context MAS	task score	**admission-time evidence check**	no	no	evidence-support only, **no evolution**	verified gists

System	Search unit	Objective	Evaluator	Learned policy	Multi-obj	Validity / artifact	Transparency
PE2 (ours)	method **vs falsifier** (minimax)	**Pareto, no weights**	**co-evolving adversary + validity gate**	**yes (cross-run)**	**yes**	**yes — real-vs-artifact, the central novelty**	**method-gene library + certificate**

Reading of the table for the rebuttal: PE2 is not "one more evolve loop." It occupies the **exact open gap** the field's own 2026 papers name but do not fill — *robustness to evaluator infidelity in a domain (wet-lab biology) where the evaluator is provably imperfect.* We can **adopt** the baselines' strong parts as engineering (SimpleTES width-scaling + IRFT credit; AutoScientists' 2σ seed-gate as a *special case* of our validity gate; DeLM's admission-time verification as a *special case* of our certificate) and still have a distinct core contribution.

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11. Ablations — answer R2-9 directly, and make "we win" mean "we generalize + don't cheat"

Headline metrics are **held-out generalization gap** and **artifact-rejection rate**, *not* in-distribution score. The anti-Goodhart ladder is the key ablation:

1. fixed scalar (current PE) →
2. ∘ Pareto (no weights) →
3. ∘ validity gate `V` →
4. ∘ adversarial falsifier co-evolution →
5. ∘ method-gene library (cumulative/transfer) →
6. ∘ learnable cross-run policy.

Each rung should **lower the held-out gap and the artifact rate** even if it *lowers or leaves flat the in-distribution score* — that inversion is the whole point and is itself the figure that refutes "you just optimized the benchmark." Plus per-component on/off (analyzer, mutator, QD-with-orthogonal-descriptors, cascade) and the external arms (AlphaEvolve/OpenEvolve/AIDE/Claude Code/human-week) under identical harness.

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12. Core loop (pseudocode)

```
def pe2_evolve(seed_methods, falsifier_generators, V, policy, budget):
    P = Archive(descriptors=[gene_composition, cost, robustness_profile])  # real orthogonal QD
    A = FalsifierArchive(quality_diversity=True, hall_of_fame=True)        # POET min-criterion
    library = MethodGeneLibrary()
    P.add_all(seed_methods); A.seed(falsifier_generators)

    while budget.remaining():
        # ---- METHOD move ----
        parent, inspirations = policy.select_parent(P, library)
        gene = policy.propose_gene(parent, library, mode=policy.explore_exploit())
        child = mutate(parent, gene)                      # analyzer→mutator, but gene-typed
        # cascade: cheap falsifiers first, promote survivors
        scores = cascade_eval(child, A, V)                # V-gated, held-out only
        fit = cvar(scores, alpha)                         # distributionally-robust fitness
        if child.is_new(content_hash):                    # dedup
            cert = certify(child, gene, A, V)             # runs the A/B ablation → certificate
            if cert.artifact_p > THRESH and cert.held_out_delta > 0:
                library.add(gene, cert)                   # cumulative, transparent
            P.insert(child, fit, cert)

        # ---- FALSIFIER move (co-evolution) ----
        if step % k == 0:
            chi = policy.propose_falsifier(A, elite=P.elite())  # target elite weaknesses
            if minimal_criterion(chi, P):                 # solvable-by-some, unsolved-by-elite
                A.insert(chi, regret=chi.regret(P.elite(), V))

        policy.credit(trajectory)                         # cross-run trajectory-level learning

    front = P.pareto_front()                              # no scalarization
    return front, library, router_from(front, A, V)       # variant router for §7
```

13. Reviewer-comment → design-feature traceability (the money table for the response letter)

Reviewer comment	PE2 feature that answers it
R1-2f / R2-8 circular eval (objective = report metric)	P1 disjointness + Primitive 1 (fitness = held-out worst-case, never the reported metric)
R1-2g "beautiful not real"	Primitive 3 validity gate + P3 artifact-rejection p-value
R1-2h black box	Primitive 2 method-gene certificate + online diff/claim viewer
R1-3 no variance / failure distribution	P5 distributional reporting + Primitive 4 (policy raises floor/shrinks variance)
R1-3 Harmony/BBKNN "more modest"	certificate explains the robustness ceiling (a result, not an omission)
R1-4 / R2-10 wrong super-human counterfactual	P4 human-week + Claude Code + AlphaEvolve/OpenEvolve/AIDE arms; "beyond human baseline" wording
R1-5 which variant to pick	§7 variant router (regime-matched, certified)
R1-min1 cross-dataset generalizability	P2 leave-condition-out (this becomes the headline design, not a caveat)
R1-min2 optimize vs explore modes	recast as explicit policy modes in Primitive 4; documented
R2-9 novelty not ablated	§11 anti-Goodhart ablation ladder (held-out gap + artifact rate)
R2-10 human-designed objective weights	Primitive 3 Pareto (no weights)

Reviewer comment	PE2 feature that answers it
R2-10 same-cells split ≠ generalization	P2 (banned)
R2-10 overcorrection / Silhouette incomplete	Primitive 3 `V` penalizes it explicitly
R2-11 embryo circularity (Cer1 boundary)	§9.3 orientation-artifact trap falsifier + `V`
R2-14 vcRouter tautology	§9.4 ground truth = real held-out downstream performance
R1-11 / R2-7 cost / time / robustness	§6.3 cascade + width-scaling + §9 cost table
R1-7 / R2-11 human-vs-agent boundary	§5.3 explicit boundary; certificate records human inputs

Every reviewer criticism of Pantheon-Evolve maps to a *designed feature*, not a promise — which is the strongest possible shape for a response letter.

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14. Honest limitations (state them; reviewers reward candor)

- **Validity-model fidelity is itself a bound.** `V` can be wrong; we mitigate with multiple ground-truth sources (sims + held-out measured genes + orthogonal assays) and report `V`'s own validation. But `V` is *strictly better than a fixed scalar* — it can only add rejection power.
- **Co-evolution can cycle/disengage** — mitigated by hall-of-fame + QD-on-falsifiers + minimal-criterion admission (§3.4); report the falsifier difficulty ladder as evidence.
- **Higher evaluation cost** than single-population evolution — paid down by cascade (§6.3), cheaper generators, and cumulative reuse from the library; reported transparently.
- **Autonomy is bounded and stated** (§5.3): human supplies question + validity sources; system discovers method + preconditions. We will *not* re-use "autonomous discovery"/"singularity" framing (also fixes R2-2, the Discussion tone).

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15. Phased implementation plan (grounded in the current repo)

Reuse the existing `pantheon/evolution/` scaffold; the changes are surgical where possible.

- **M0 — De-risk the two falsifiers first (Weeks 1–3).** On batch correction only: (a) implement P1/P2 harness (leave-study-out; objective/report disjointness) and re-run the *existing* Scanorama evolution under it — this alone tells us how much of the current headline survives an honest split (critical to know before the rebuttal). (b) Implement one trap (bio-null) + a v0 validity gate. Deliverable: the anti-Goodhart ladder rungs 1→3 on one task.
- **M1 — Minimax + certificate (Weeks 3–6).** Add the falsifier archive, CVaR fitness, method-gene records + auto-ablation certificate. Wire real orthogonal QD descriptors; bound & dedup the archive; reinstate cascade. `config.py`: retire dead config, add falsifier/validity knobs. Deliverable: ladder rungs 4–5 on batch correction, with distribution over seeds (R1-3).

- **M2 — Learnable policy + variant router (Weeks 6–9).** Cross-run trajectory credit; regime router (§7). Deliverable: ladder rung 6; learned-policy arm Pareto-dominates fixed-search arm across seeds.
- **M3 — Port the template to the other three tasks (Weeks 9–14).** Gene panel, 3D reconstruction, virtual-cell downstream — each is *just* new falsifier generators + a `V` head; the loop is unchanged (that uniformity is the platform argument). Cross-domain method-gene transfer experiment (§4.2) is the flagship "method invention" figure.
- **Throughout — counterfactual arms (P4):** run Claude Code + AlphaEvolve/OpenEvolve/AIDE + an expert-week arm under the identical harness. This is the single most persuasive addition for both reviewers and is mostly harness/plumbing, not new algorithm.

Contamination guardrail (from `benchmarks/README.md`): tasks used to *evolve*/*tune* must be disjoint from tasks *reported*; use public held-out/private splits; OmicBench is in-house → dev/supplementary only, never the neutrality argument.